National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Faculty of Computing

CS 368: Reinforcement Learning

Lab 12 & 13: Solving MuJoCo Environment Using RL

(Open Ended Lab)

Date: 19 December 2025

Time: 02:00 PM – 5:00 PM

Instructor: Dr. Zuhair Zafar
Lab Engineer: Hafiz Arslan Ramzan

Lab 12 &13: Solving MuJoCo Environment Using Reinforcement Learning

Objectives

The goal of this lab is to apply reinforcement learning (RL) algorithms to solve the Hopper MuJoCo environment from the Gymnasium library.

Tools/Software Requirement:

Google Colab, Python, Gymnasium Library

Introduction

MuJoCo stands for Multi-Joint dynamics with Contact. It is a physics engine for facilitating research and development in robotics, biomechanics, graphics and animation, and other areas where fast and accurate simulation is needed. There is physical contact between robots and their environment - and MuJoCo attempts at getting realistic physics simulations for the possible physical contact dynamics by aiming for physical accuracy and computational efficiency.

In this lab, we focus on the Hopper environment, a continuous-control task from the MuJoCo suite where the objective is to train a single-legged robot to hop forward efficiently and stably. Hopper presents notable challenges for reinforcement learning algorithms due to its complex dynamics, continuous action space, and the need for balance, coordination, and smooth motion control. These characteristics make it an excellent benchmark for evaluating the performance and robustness of modern RL methods.

Through this lab, you will gain hands-on experience with implementing RL algorithms, hyper-parameter tuning, and optimizing agent performance.

Lab Tasks

The task of this lab is to develop, train, and evaluate a reinforcement learning agent to solve the **Hopper** from MuJoCo environment using the Gymnasium library. You will preprocess the environment, choose and implement an RL algorithm, optimize hyperparameters, and analyze the agent's performance through experimentation and the plots of learning curves and the returns per time steps.

Deliverable:
Please submit your notebook on LMS before the deadline **(24th December 2025, 11:59pm).**

Lab Rubrics

Assessment	Does not meet expectation (2/4 marks)	Meets expectation (6/8 marks)	Exceeds expectation (10 marks)
Software Problem Realization **(CLO1, PLO1)**	The student struggles to formulate the problem as RL and does not apply RL algorithm to solve it. There is a lack of understanding of the problem's requirements and no attempt to solve the MuJoCo enironment effectively.	The student formulates the problem as RL with some guidance, applies appropriate RL algorithm with hints, and shows it's working. However, the approach might not be fully optimized or lacks a thorough justification.	The student independently formulates the given problem as RL, applies RL algorithm using SOTA libraries without guidance, and effectively solves the MuJoCo environment. The approach is fully optimized and can be applied to different MuJoCo environments.
Software Tool Usage **(CLO4, PLO5)**	Code has syntax errors, and the implementation of the RL algorithm is incorrect or incomplete. The code is not modular and lacks comments for readability and reuse. The student shows limited ability to use gymnasium / pytorch / SOTA library functions where required.	The student has implemented the RL algorithm correctly for the given problem with minor mistakes. The code is mostly correct in terms of syntax and functionality but might not be optimized or well-structured for reuse. Some documentation is provided. The student also shows working knowledge of gymnasium and SOTA library where required.	The student has implemented the RL algorithm using SOTA library efficiently and correctly. The code is clean, modular, well-documented, and follows best practices for reproducibility and reuse. The student demonstrates full command of the gymnasium and SOTA library and its functions.